



CS-404 Big Data Analytics ASSIGNMENT 03

ETL Pipeline & Analytics

BDA Spring 2026 By: Ms. Zahida Kausar (Total Marks: 30)

Submission deadline: Wednesday 6th May, 2026 (11:59PM)

Overview

Building directly on your A2 (Assignment02) ingestion and profiling work, this milestone takes your cleaned data through the complete data warehousing lifecycle: ETL transformation, analytical querying, and business insight generation. All work must use the same dataset approved in A2.

Learning Outcomes

- Implement a PySpark ETL pipeline that transforms and loads data into a structured warehouse
- Write Spark SQL queries to answer real business questions
- Generate and interpret data visualizations from warehouse data
- Optimize pipeline performance using industry-standard techniques

Task 1: PySpark ETL Pipeline (10 marks)

Implement etl.py that picks up from where A2 left off, your data is already ingested and cleaned. This script handles transformation, modelling, loading, and validation only:

Step	Action
Transform	Derive new columns (e.g., revenue totals, time-of-day flags, age buckets), normalize categorical values, standardize date/time formats
Load	Write each table to HDFS as Parquet e.g. /warehouse/processed/fact_sales/year=2026/
Validate	Verify row counts, assert no nulls in key columns, log a per-table summary of records written

Important: Every transformation must reference the relevant A2 profiling finding in an inline comment. Undocumented transformations receive zero marks.

Task 2: Analytical Queries & Visualizations (14 marks)

Using Spark SQL on your loaded Parquet warehouse, answer at least 5 business questions from A2. Then visualize your findings:

Queries (8 marks)

Requirement	Detail
5 Spark SQL queries	One per business question, with explanatory comments
Window functions	At least 2 queries must use RANK(), ROW_NUMBER(), or LAG()
Time-based analysis	At least 1 query must show a monthly, quarterly, or yearly trend
Business interpretation	3–5 sentences per query explaining what the result means and what action a business could take

Visualizations (6 marks)



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Department of Computing

Produce 4 charts from your warehouse data, one of each type:

- Line or area chart (trend over time)
- Bar or grouped bar chart (category comparison)
- Heatmap or scatter plot (correlation between numeric facts)
- Summary dashboard (at least 3 subplots in one figure)

Each chart must have a title, labeled axes, and a 2–3 sentence interpretation.

Task 3: Pipeline Optimization & Final Report (6 marks)

Optimization (3 marks)

Apply and document at least 2 of the following techniques:

- **Partitioning** — partition Parquet output by a meaningful column
- **Caching** — use `.cache()` or `.persist()` on frequently reused DataFrames
- **Broadcast joins** — apply `broadcast()` when joining small dimension tables to the fact table
- **Query plan analysis** — show `.explain(True)` output for one complex query and interpret it

Include a code snippet and a brief performance explanation for each technique used.

Final Report (3 marks)

Submit `final_report.pdf` covering:

Section	Requirements
A2 Traceability Table	Every A2 data quality issue → transformation applied → PySpark function used
Row Count Verification	Raw vs. post-ETL counts for each table, discrepancies explained
Query Results & Insights	All 5 query outputs with business interpretations
Visualizations	All 4 charts embedded with captions and interpretations
HDFS Screenshot	/warehouse/processed/ directory showing Parquet partitions

Submission Requirements

File	Description
<code>etl.py</code>	PySpark transformation, modelling, loading, and validation script
<code>analytics.py</code>	Spark SQL queries and visualization script
<code>final_report.pdf</code>	Complete report covering all 3 tasks
<code>hdfs_screenshot.png</code>	/warehouse/processed/ directory screenshot
<code>requirements.txt</code>	All dependencies
<code>README.md</code>	Setup instructions, how to run both scripts, group members

File naming: GroupNumber_A3_BDA.zip

P.S. Please begin working on this assignment today. No deadline extensions will be granted due to the semester project presentations scheduled for the final week.